



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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March 25, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- SpaceX has launch the satellites with a concept of reusable booster, the “Falcon 9/Falcon 9 Heavy”.
- The main scope of this is the significant reduction in the cost per kg.
- Falcon 9 successful recovery depends on the following features:
 - Orbit
 - Booster Version
 - Launching Sites
 - Payload Mass
- With these features, the Machine Learning supervised Classification Model developed predicted

Introduction

- Falcon 9 rocket launches with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, the reason of the savings of a great amount of million dollars is due to that Space X can reuse the 1st stage. With this, we can determine that if the first stage will land, we can determine the cost of a launch.
- In this capstone, we will predict if the Falcon 9 first stage will land successfully using the dataset from Falcon 9 rocket launches advertised in the website.

Section 1

Methodology

Methodology

Executive Summary

1. Data collection methodology:
2. Perform data wrangling
3. Perform exploratory data analysis (EDA) using visualization and SQL
4. Perform interactive visual analytics using Folium and Plotly Dash
5. Perform predictive analysis using classification models

Data Collection – SpaceX API

- Data collected using REST API from SpaceX by making a GET request to the API and then decoded the response content as a JSON result which was converted into Pandas Dataframe.
- <https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

Task 1: Request and parse the SpaceX launch data using the GET request

To make the requested JSON results more consistent, we will use the following static response object for this project:

```
[9]: static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_ap
```

We should see that the request was successfull with the 200 status response code

```
[10]: response=requests.get(static_json_url)
```

```
[11]: response.status_code
```

```
[11]: 200
```

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
[18]: # Use json_normalize meethod to convert the json result into a dataframe
data = pd.json_normalize(response.json())
```

Data Collection - Scraping

- Perform Web Scraping from Falcon 9 launch records from Wikipedia using BeautifulSoup and a GET Request to extract the records from the HTML table.
- https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/web_scraping.ipynb

```
[4]: static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"
```

Next, request the HTML page from the above URL and get a `response` object

TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

```
[5]: # use requests.get() method with the provided static_url
     # assign the response to a object
```

```
import requests
```

```
response = requests.get(static_url).text
```

Create a `BeautifulSoup` object from the HTML `response`

```
[6]: # Use BeautifulSoup() to create a BeautifulSoup object from a response text content
     soup = BeautifulSoup(response, 'html5lib')
```

Print the page title to verify if the `BeautifulSoup` object was created properly

Data Wrangling

- After creating the Pandas Dataframe from the collected data, the dataset was filtered using the Booster-Version column to keep only the data related to the Falcon 9 launches.
- Also, Payload Mass column had some missing values that were replaced with the mean of the column.
- <https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

TASK 4: Create a landing outcome label from Outcome column

Using the `Outcome`, create a list where the element is zero if the corresponding row in `Outcome` is in the set `bad_outcome`; otherwise, it's one. Then assign it to the variable `landing_class`:

```
[18]: # landing_class = 0 if bad_outcome
      # landing_class = 1 otherwise
      landing_class = [0 if outcome in bad_outcomes else 1 for outcome in df['Outcome']]
```

This variable will represent the classification variable that represents the outcome of each launch. If the value is zero, the first stage did not land successfully; one means the first stage landed Successfully

```
[21]: df['Class']=landing_class
      df[['Class']].head(8)
```

```
[21]:
```

	Class
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	1

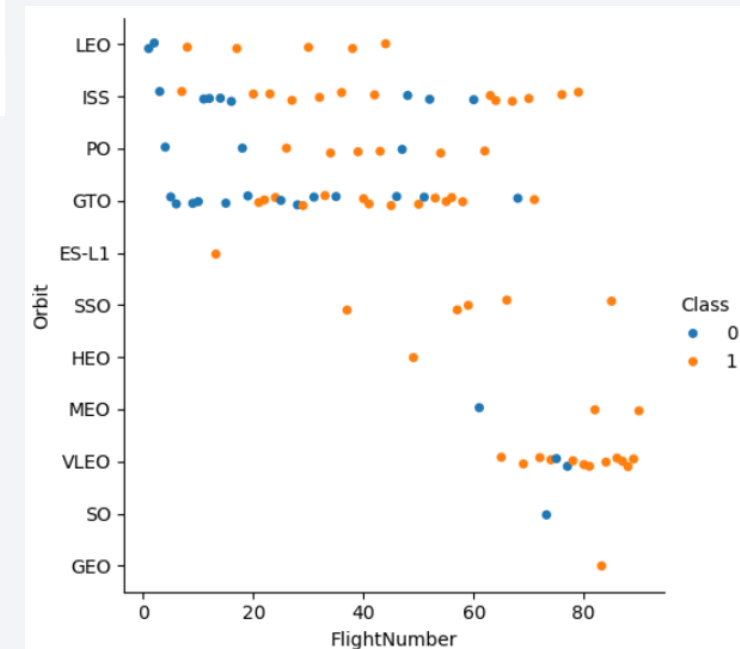
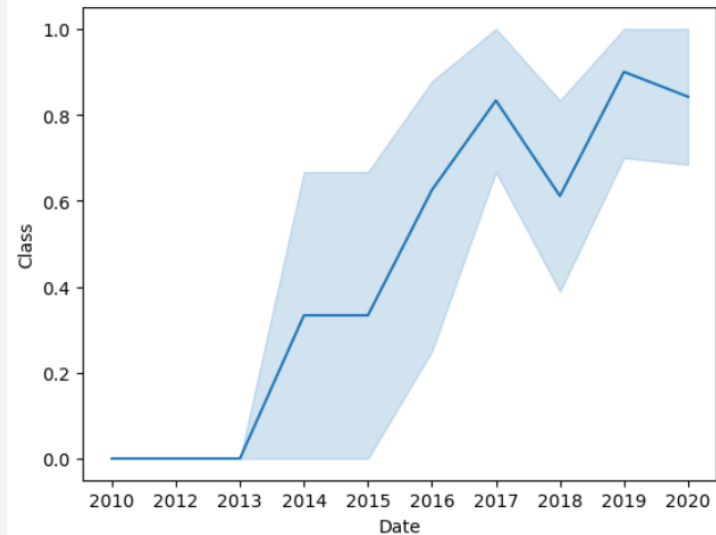
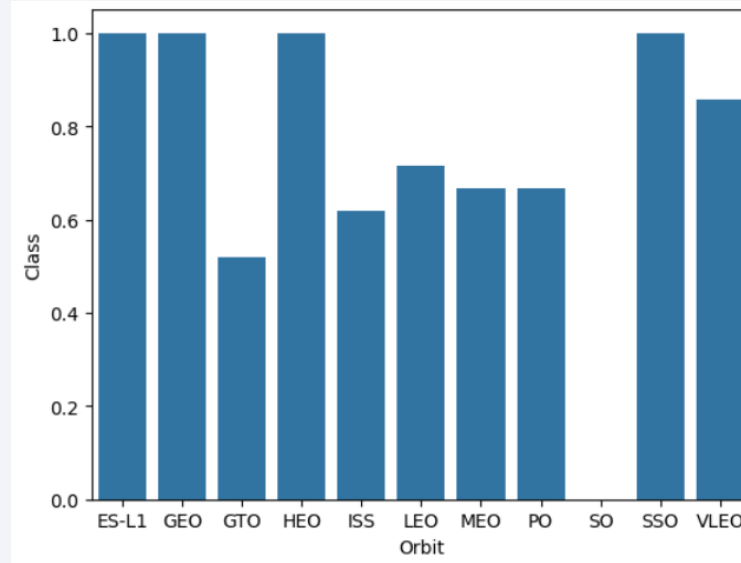
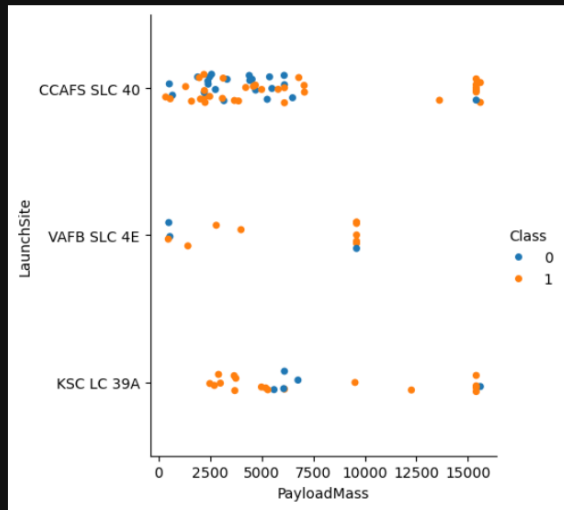
EDA with Data Visualization

- EDA performed and Feature Engineering using Pandas and Matplotlib libraries.
 1. EDA.
 2. Prepare Data Feature Engineering.
 3. Visualize the relationship between Flight Number and Launch Site, Payload Mass and Launch Site, Flight Number and Orbit, Payload Mass and Orbit using Scatter Plots.
 4. Bar Chart used to visualize the relationship between Success rate of each Orbit.
 5. Line Plot used to visualize the Launch site Success yearly trend.
- <https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/edadataviz.ipynb>

EDA Visualization Charts

```
[7]: # Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis  
sns.catplot(y='LaunchSite', x='PayloadMass', hue='Class', data=df)
```

```
[7]: <seaborn.axisgrid.FacetGrid at 0x689e510>
```



EDA with SQL

https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

The following SQL queries were performed:

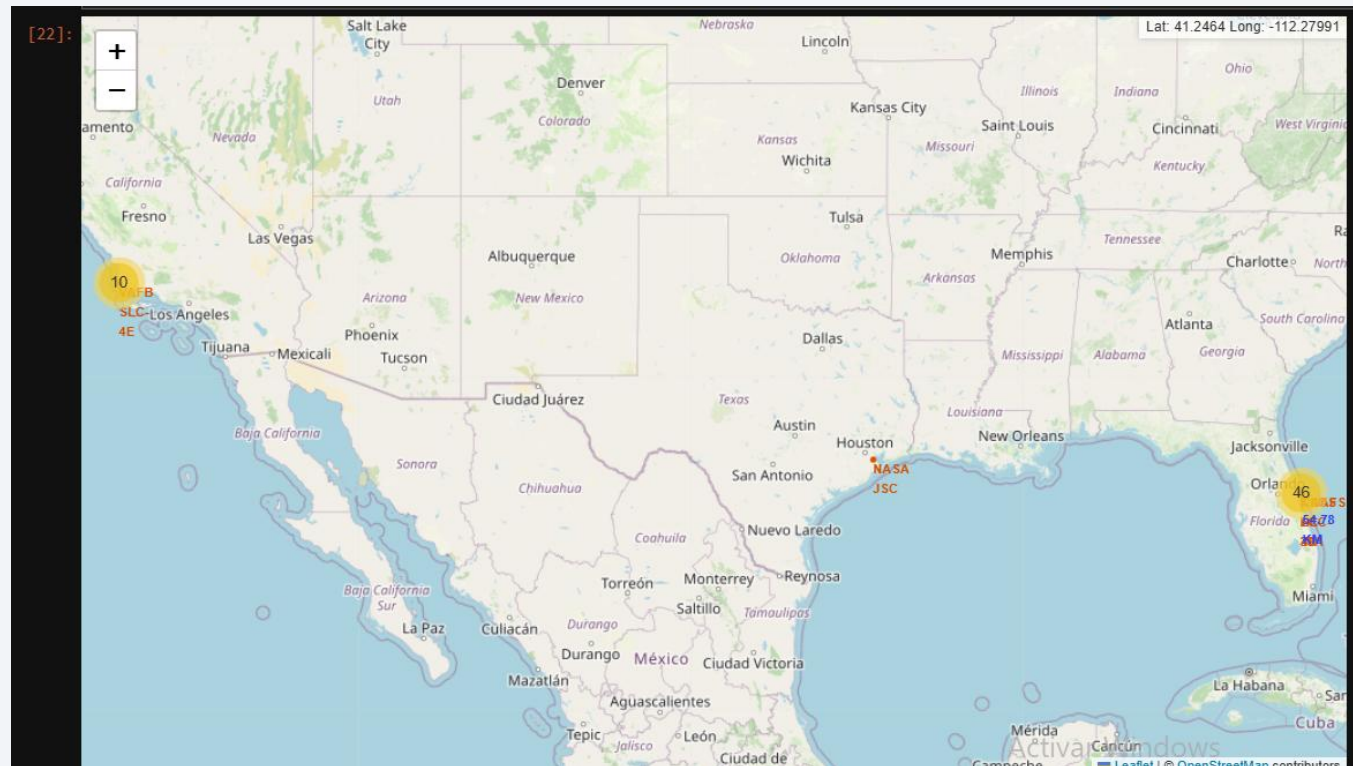
- **Display the names of the unique launch sites in the space mission**
 - %sql select * from SPACEXTABLE limit 1;
- **Display 5 records where launch sites begin with the string 'CCA'**
 - %sql select * from SPACEXTABLE where Launch_Site like "CCA%" limit 5;
- **Display the total payload mass carried by boosters launched by NASA (CRS)**
 - %sql select sum(PAYLOAD_MASS__KG_) as TOTAL_PAYLOAD_MASS_KG, Customer from SPACEXTABLE where Customer="NASA (CRS)";
- **Display average payload mass carried by booster version F9 v1.1**
 - %sql select Booster_Version, avg(PAYLOAD_MASS__KG_) as TOTAL_PAYLOAD_MASS_KG from SPACEXTABLE where Booster_Version = "F9 v1.1";
- **List the date when the first succesful landing outcome in ground pad was acheived.**
 - %sql select min(Date), Landing_Outcome from SPACEXTABLE where Landing_Outcome='Success (ground pad)';

EDA with SQL

- **List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000**
 - `%sql select Booster_Version, PAYLOAD_MASS__KG_, Landing_Outcome from SPACEXTABLE where Landing_Outcome='Success (drone ship)' and PAYLOAD_MASS__KG_ between 4001 and 5999;`
- **List the total number of successful and failure mission outcomes**
 - `%sql select Mission_Outcome, count(Mission_Outcome) from SPACEXTABLE group by Mission_Outcome;`
- **List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.**
 - `%sql select distinct Booster_Version from SPACEXTABLE where (select max(PAYLOAD_MASS__KG_) from SPACEXTABLE);`
- **List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.**
 - `%%sql select substr(Date,6,2), Booster_Version, Launch_Site, Landing_Outcome from SPACEXTABLE where Landing_Outcome='Failure (drone ship)' and substr(Date,0,5)='2015';`
- **Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.**[1](#)
 - `%%sql select Date, Landing_Outcome, count(Landing_Outcome) as Total from SPACEXTABLE where Date between '2010-06-04' and '2017-03-20' group by Landing_Outcome order by Total desc;`

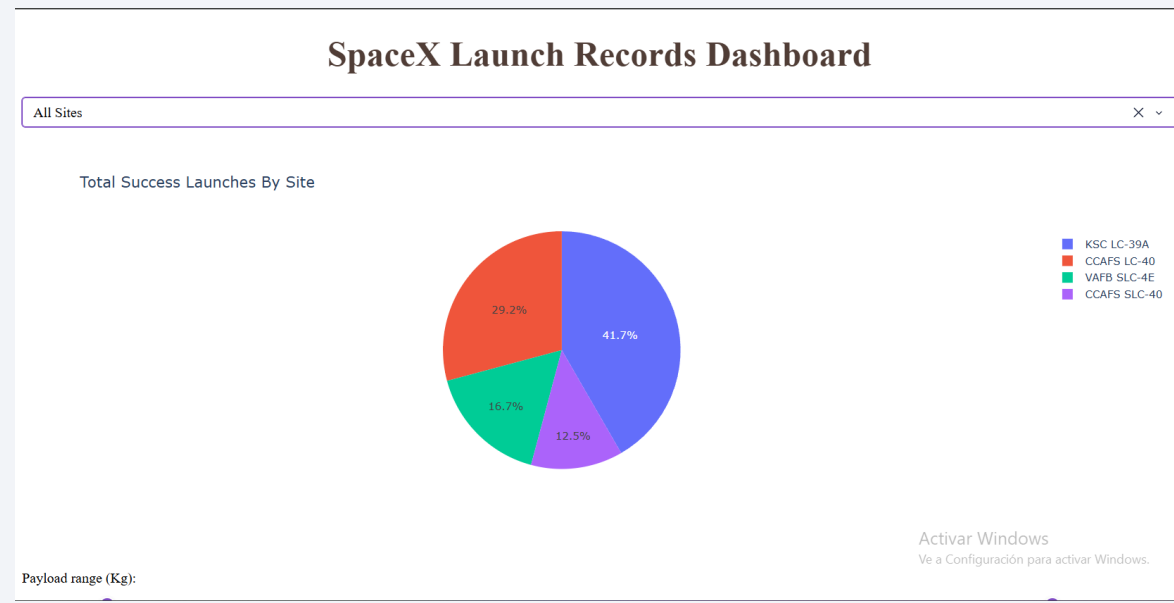
Build an Interactive Map with Folium

- Map created with the Folium library to mark all the Launch Sites along with markers, circles and lines to mark the launch failure or success for each Launch Site.
- The Launch Site outcome sets 0 for failure and 1 for success launch.
- <https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/Launch%20Sites%20Locations%20Analysis%20with%20Folium.ipynb>



Build a Dashboard with Plotly Dash

- Interactive Dashboard Application built with Plotly Dash to:
 - Add Launch Site Dropdown input component.
 - Add a Callback function to render 'success-pie-chart' based on the selected Launch Site.
 - Add a Range slider to select the Payload.
 - Add a Callback function to render 'success-payload-scatter' chart.
- <https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/spacex-dash-app.py>



Predictive Analysis (Classification)

- After the data was loaded as Dataframe and the EDA was performed, Training labels were determined:
 - Create Numpy array from the Class column and assigned it to the variable Y as the outcome variable.
 - Standardize the feature dataset (X) by transforming it using preprocessing StandardScaler method from Sklearn library.
 - Split data into training and testing subsets using the train_test_split method with test size parameter 0.2 and random state 2.

Predictive Analysis (Classification)

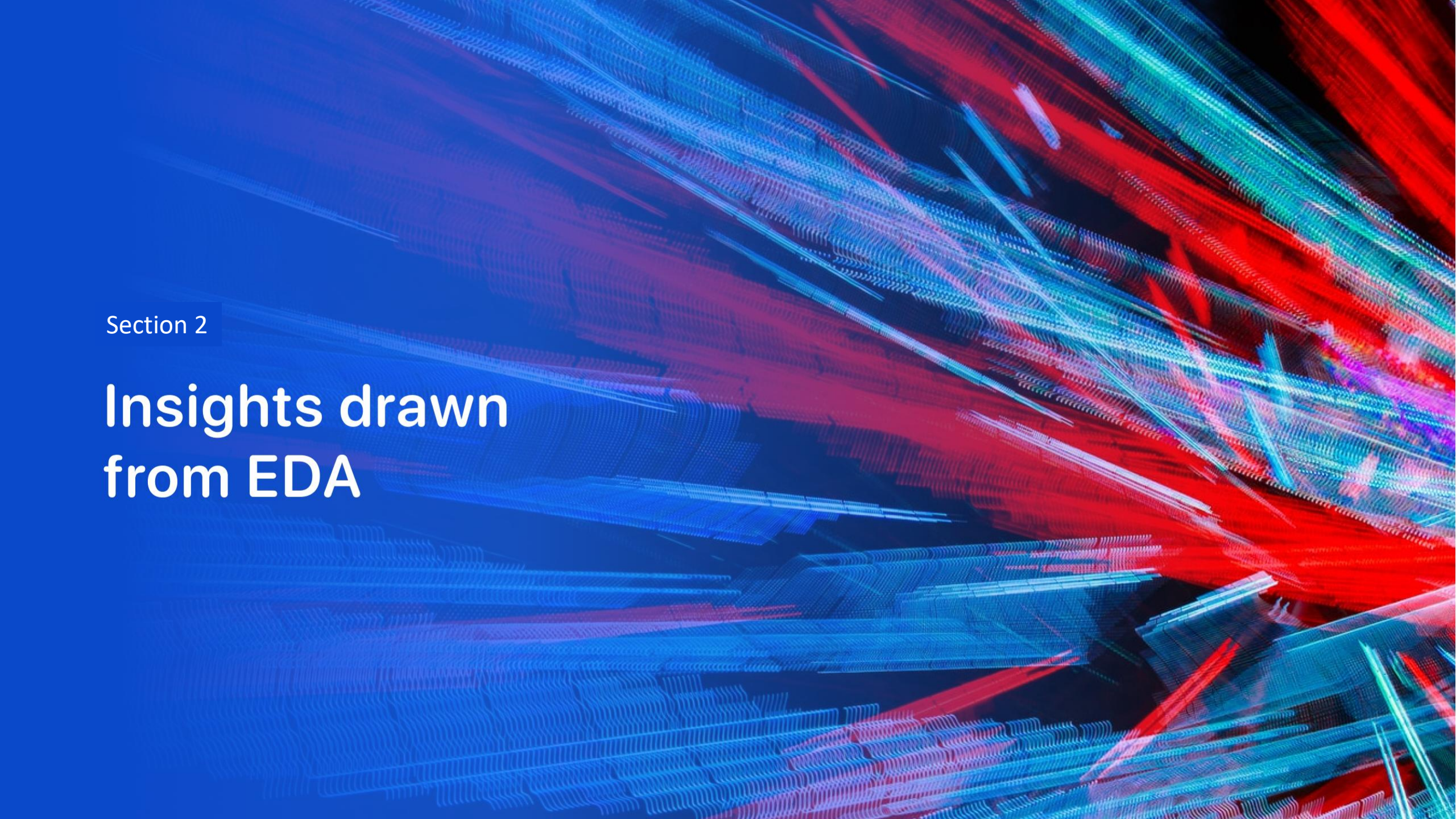
- Then, to find the best Machine Learning Model that would perform best, we use Test Data to create and predict with Classification Tree, SVM, KNN and Logistic Regression models:
 - Create an object for all the algorithms to be used and also create a GridSearchCV object with the assigned set of parameters for each model.
 - The GridSearchCV object was created for cv=10 and fit the training data to find the best hyperparameter and accuracy.
 - When fitting the training set, the best parameters were displayed with the 'best_params_' and the accuracy with the 'best_score_'.
 - Method '.score' to calculate the accuracy of test dataset for each model and plotted a confusion matrix for each model using the test and predicted outcomes.

0	
Method	Test Data Accuracy
Logistic_Reg	0.833333
SVM	0.833333
Decision Tree	0.777778
KNN	0.833333

- [https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/SpaceX Machine%20Learning%20Prediction Part 5.ipynb](https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/SpaceX%20Machine%20Learning%20Prediction%20Part%205.ipynb)

Results

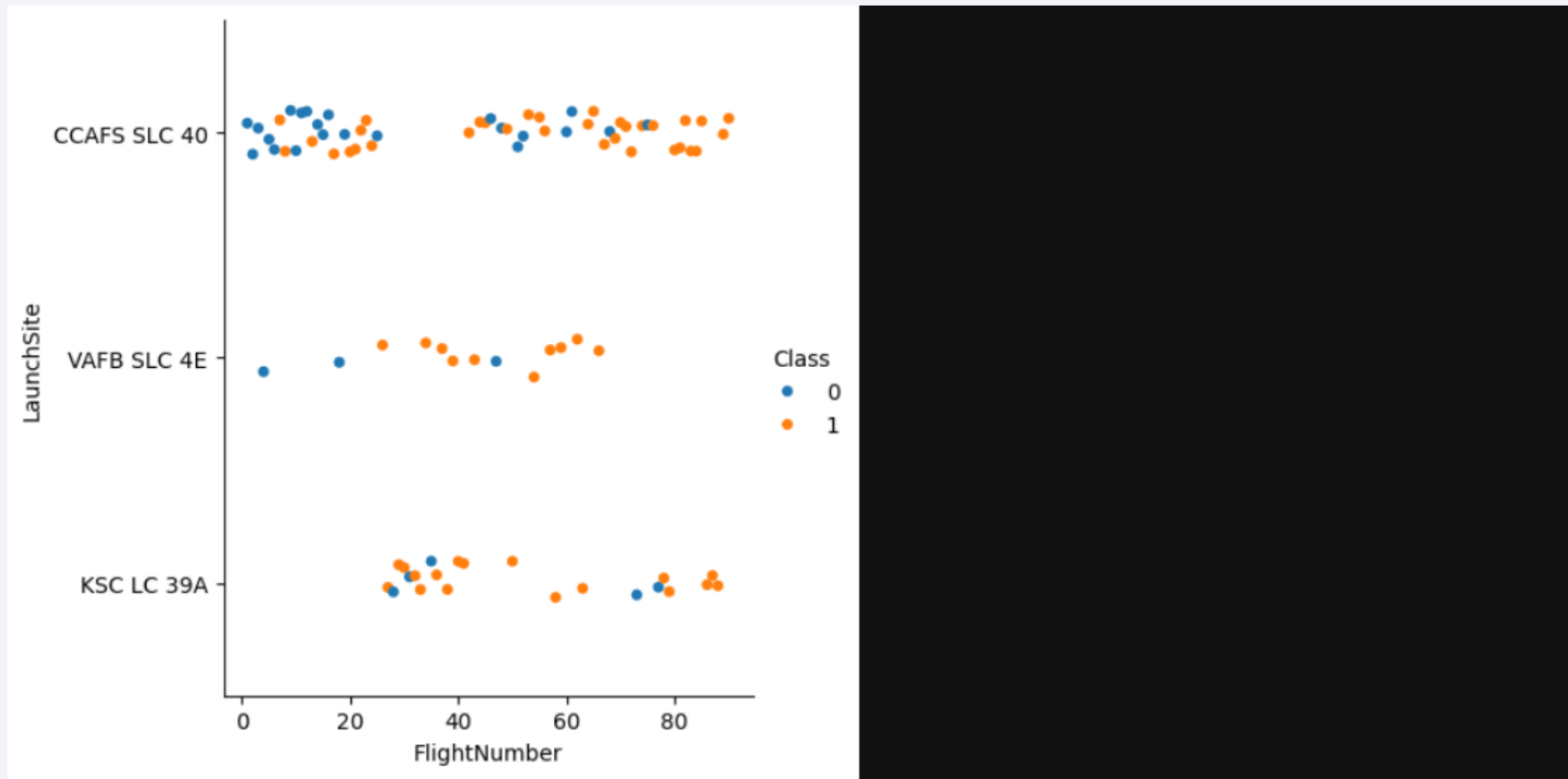
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

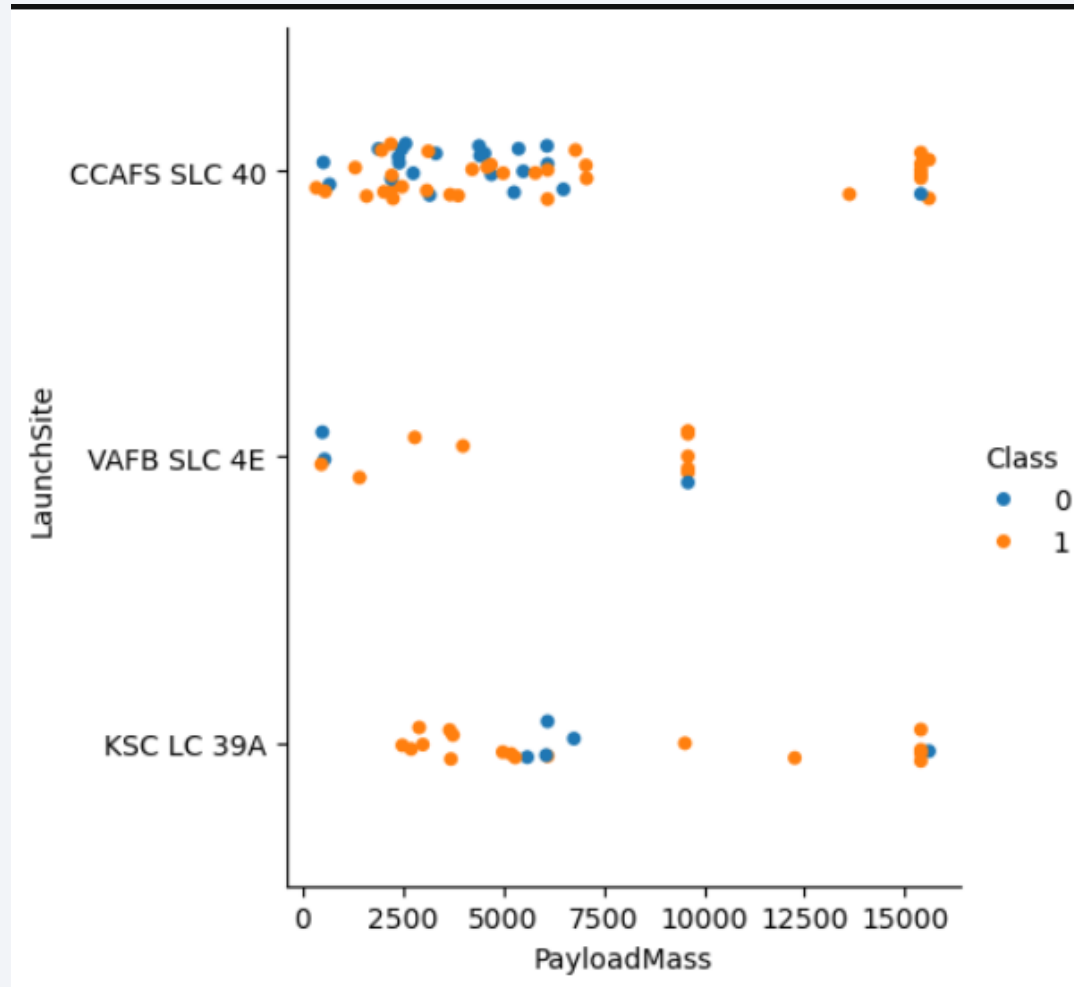
Insights drawn from EDA

Flight Number vs. Launch Site



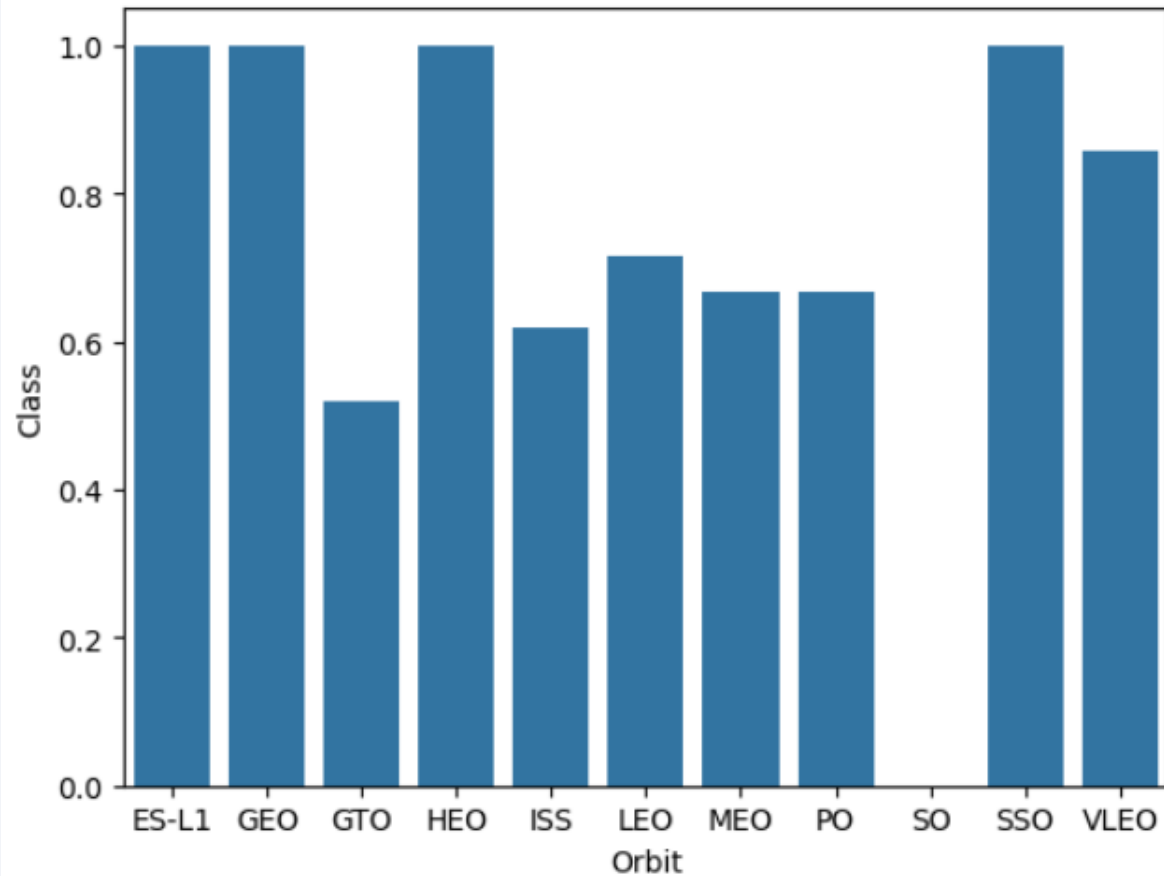
Now try to explain the patterns you found in the Flight Number vs. Launch Site scatter point plots. They started testing with CCAFS SLC 40 earlier and several have failure and had a gap of 20 flights, where there the testing as in other launchsites, but after that gap, the results where more succesful. For BAFB SLC 4E, they tested a lower amount of flights but had a better succesful rate. For KSC LC 39A, started testing in the gap for launch site CCAFS SLC 40, and they had a good succesful rate, but after that, they had a periodically test because they focused again in the CCAFS SLC 40 launch site.

Payload vs. Launch Site



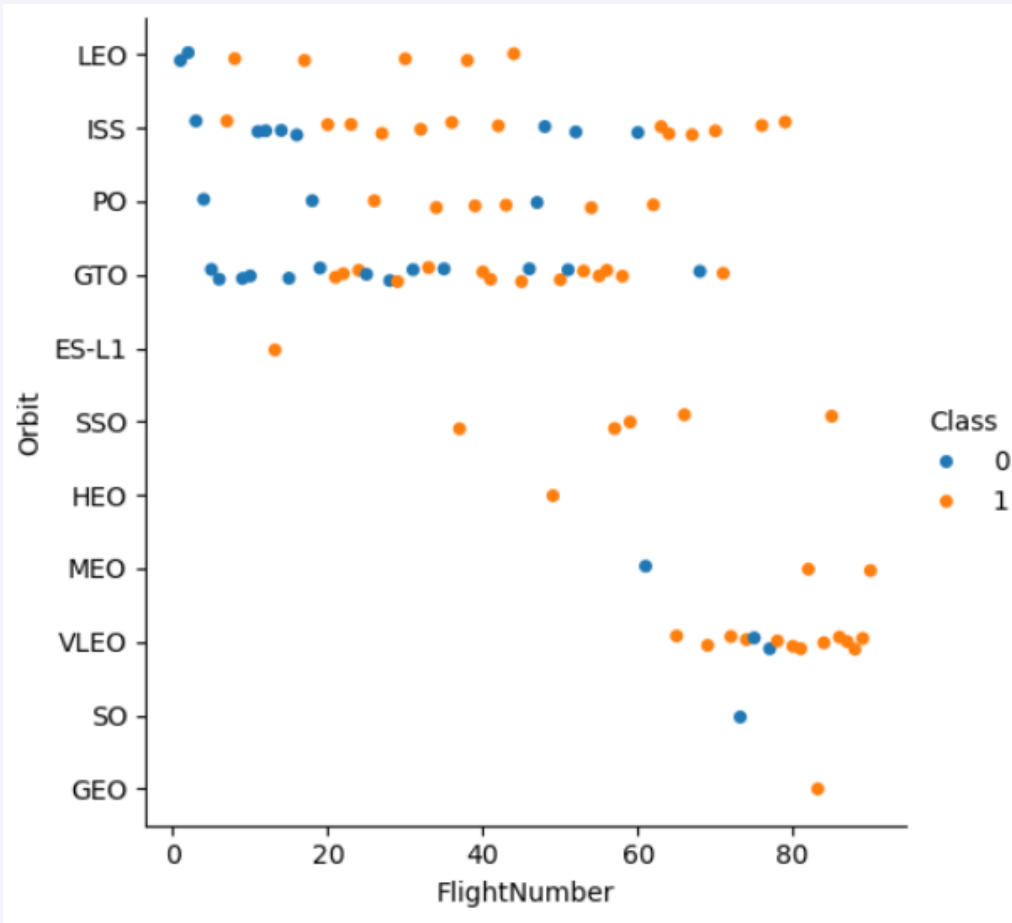
Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavy payload mass(greater than 10000).

Success Rate vs. Orbit Type



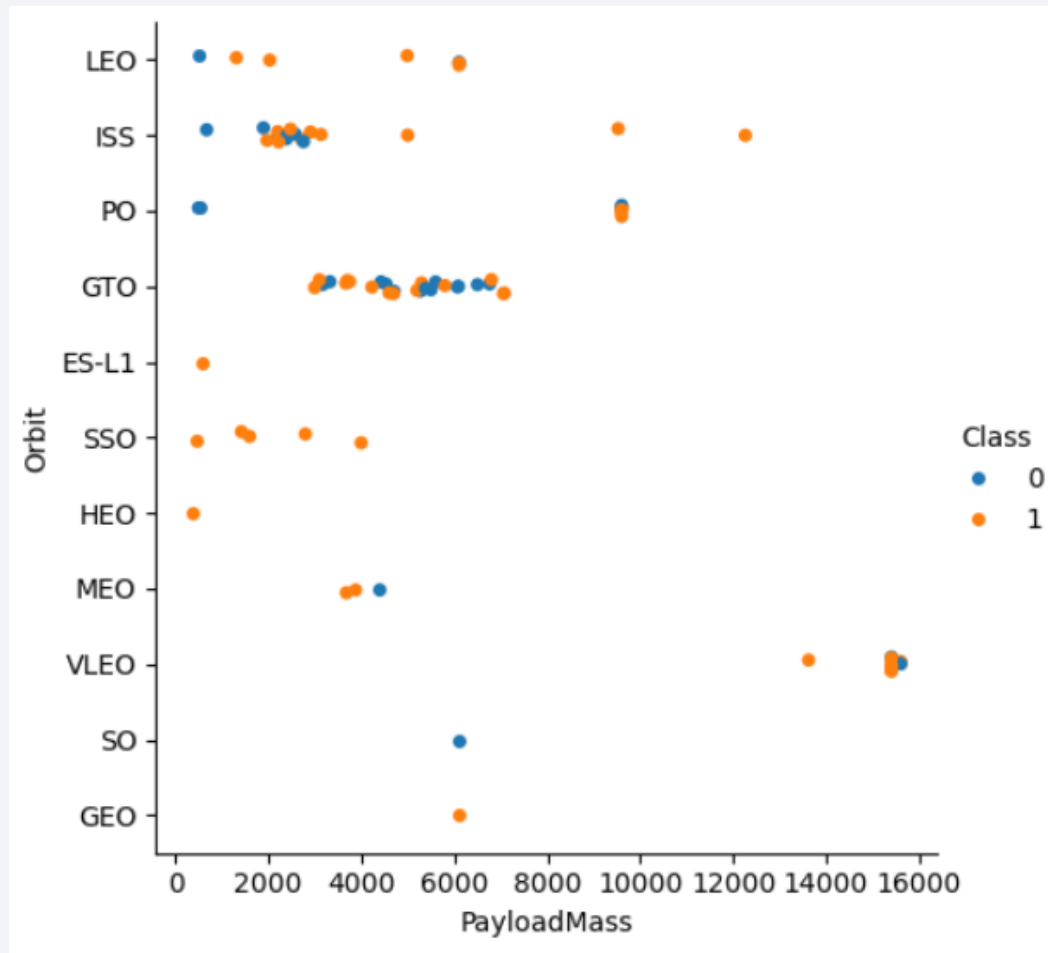
Analyze the plotted bar chart to identify which orbits have the highest success rates. Orbits ES-L1, GEO, HEO and SSO are the successful orbits having 100% success rate while orbit SO has 0% of success rate. For orbits GTO, ISS, LEO, MEO, PO and VLEO the success rate is between 50% to 85%

Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

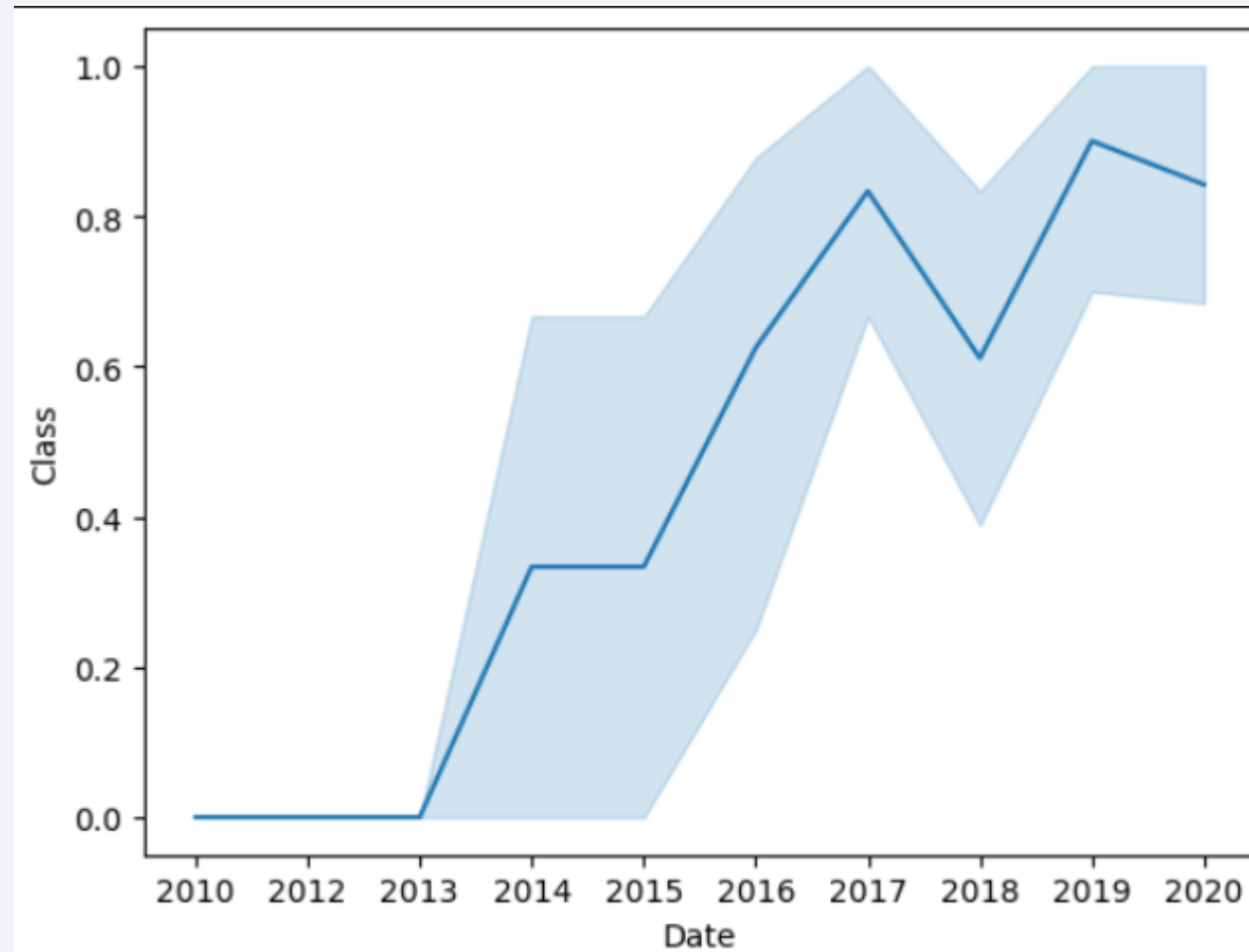
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



you can observe that the sucess rate since 2013 kept increasing till 2020

All Launch Site Names

- Using Distinct in the query, we get all the unique different Launch Sites.

```
[29]: %sql select distinct Launch_Site from SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[29]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

- Find the list of Launch Sites that starts with the prefix 'CCA', we use the LIKE operator in the query, the '%' sign in the prefix to search for the rest of the values and LIMIT to obtain the top 5 results.

```
[17]: %sql select * from SPACEXTABLE where Launch_Site like "CCA%" limit 5;
```

* sqlite:///my_data1.db
Done.

[17]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total Payload Mass with the SUM() operator and filter the data with Customer set as 'NASA (CRS)'.

```
[24]: %sql select sum(PAYLOAD_MASS_KG_) as TOTAL_PAYLOAD_MASS_KG, Customer from SPACEXTABLE where Customer="NASA (CRS)";
```

```
* sqlite:///my_data1.db  
Done.
```

```
[24]: TOTAL_PAYLOAD_MASS_KG  Customer  
-----  
45596  NASA (CRS)
```

Average Payload Mass by F9 v1.1

- Calculate the Payload Mass average with the AVG() operator in the query and then filter the results with Booster Version set as 'F9 v1.1'.

```
%sql select Booster_Version, avg(PAYLOAD_MASS_KG_) as TOTAL_PAYLOAD_MASS_KG from SPACEXTABLE where Booster_Version = "F9 v1.1";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version	TOTAL_PAYLOAD_MASS_KG
F9 v1.1	2928.4

First Successful Ground Landing Date

- Use min() to Date column to find the first landing and then filter the results with the Landing Outcome set as 'Success (ground pad)'.

```
%sql select min(Date), Landing_Outcome from SPACEXTABLE where Landing_Outcome='Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

min(Date)	Landing_Outcome
-----------	-----------------

2015-12-22	Success (ground pad)
------------	----------------------

Successful Drone Ship Landing with Payload between 4000 and 6000

- Filter the query results with Landing Outcome feature set as 'Success (drone ship)' value and the Payload Mass with the BETWEEN operator with 4,000 and 6,000 values.
 - %sql select Booster_Version, PAYLOAD_MASS__KG_, Landing_Outcome from SPACEXTABLE where Landing_Outcome='Success (drone ship)' and PAYLOAD_MASS__KG_ between 4001 and 5999;

Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)

Total Number of Successful and Failure Mission Outcomes

- Use COUNT() operator with Mission Outcome column to obtain the count of the Mission Outcomes and then GROUP BY the results with the column Mission Outcome.

```
%sql select Mission_Outcome, count(Mission_Outcome) from SPACEXTABLE group by Mission_Outcome;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	count(Mission_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Use Distinct in query for Booster Version and then use a nested query to filter the results with MAX() operator for Payload Mass column.

```
%sql select distinct Booster_Version from SPACEXTABLE where (select max(PAYLOAD_MASS_KG_) from SPACEXTABLE)
* sqlite:///my_data1.db
Done.
Booster_Version
F9 v1.0 B0003
F9 v1.0 B0004
F9 v1.0 B0005
F9 v1.0 B0006
F9 v1.0 B0007
F9 v1.1 B1003
F9 v1.1
F9 v1.1 B1011
F9 v1.1 B1010
F9 v1.1 B1012
F9 v1.1 B1013
F9 v1.1 B1014
F9 v1.1 B1015
```

2015 Launch Records

- Use substr to get the Months and year to obtain the Months, Booster version, Launch Site and Landing Outcome with 'Failure (drone ship)' value for the year 2015.

```
%%sql
select substr(Date,6,2), Booster_Version, Launch_Site, Landing_Outcome from SPACEXTABLE
where Landing_Outcome='Failure (drone ship)' and substr(Date,0,5)='2015';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

substr(Date,6,2)	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Use COUNT() operator for Landing Outcome and filter with the dates the results and then sort the results descendent way to obtain the ranking.

```
%%sql
select Date,Landing_Outcome,count(Landing_Outcome) as Total from SPACEXTABLE
where Date between '2010-06-04' and '2017-03-20' group by Landing_Outcome order by Total desc;
```

```
* sqlite:///my_data1.db
Done.
```

Date	Landing_Outcome	Total
2012-05-22	No attempt	10
2016-04-08	Success (drone ship)	5
2015-01-10	Failure (drone ship)	5
2015-12-22	Success (ground pad)	3
2014-04-18	Controlled (ocean)	3
2013-09-29	Uncontrolled (ocean)	2
2010-06-04	Failure (parachute)	2
2015-06-28	Precluded (drone ship)	1

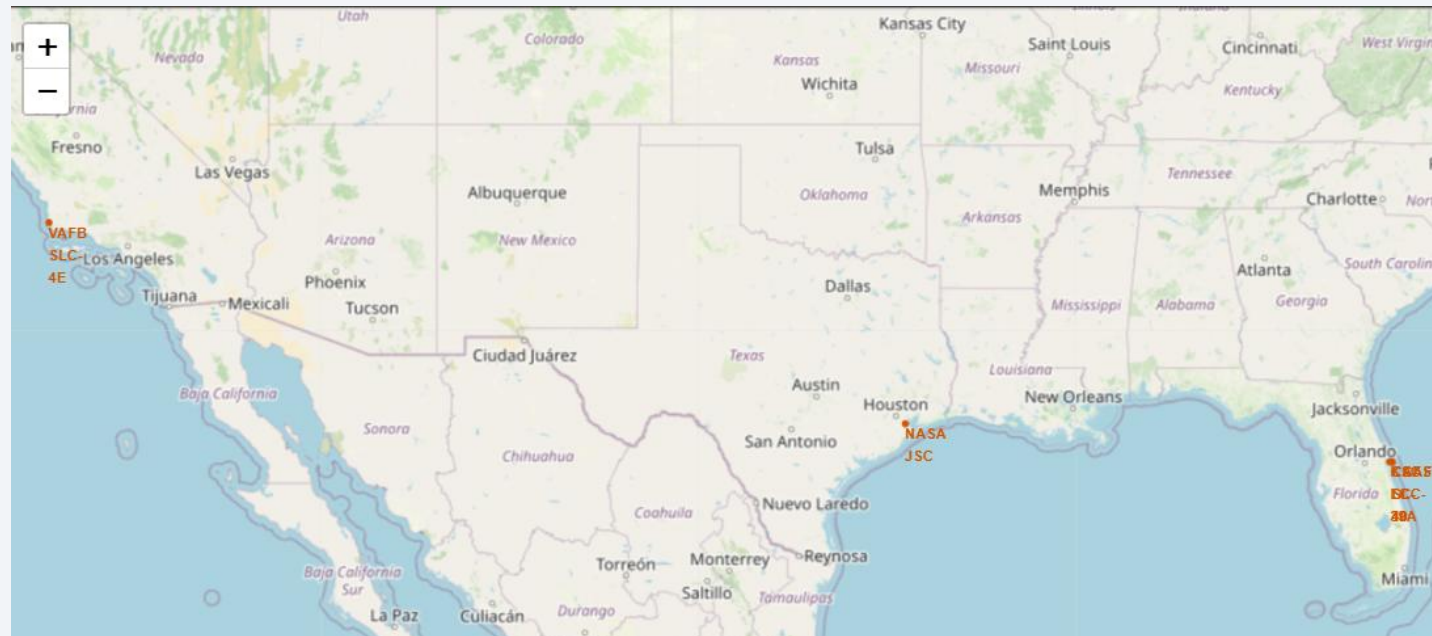
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

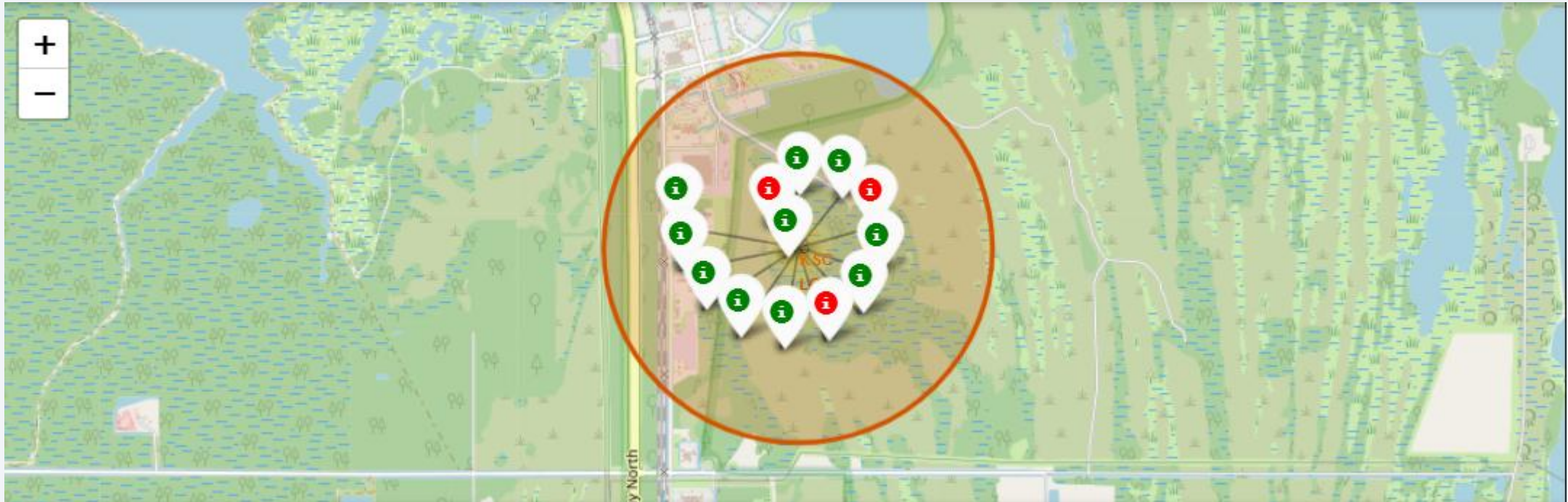
Folium Map – Launch Sites

- There are 4 Launch Sites of SpaceX:
 - VAFB SLC-4E – Vandenberg Space Launch Complex (CA)
 - KSC LC29A – Kennedy Space Center (FL)
 - CCAFS LC40 – Cape Canaveral Launch Complex 40 (FL)
 - CCAF SLC40 – Cape Canaveral Space Launch Complex 40 (FL).



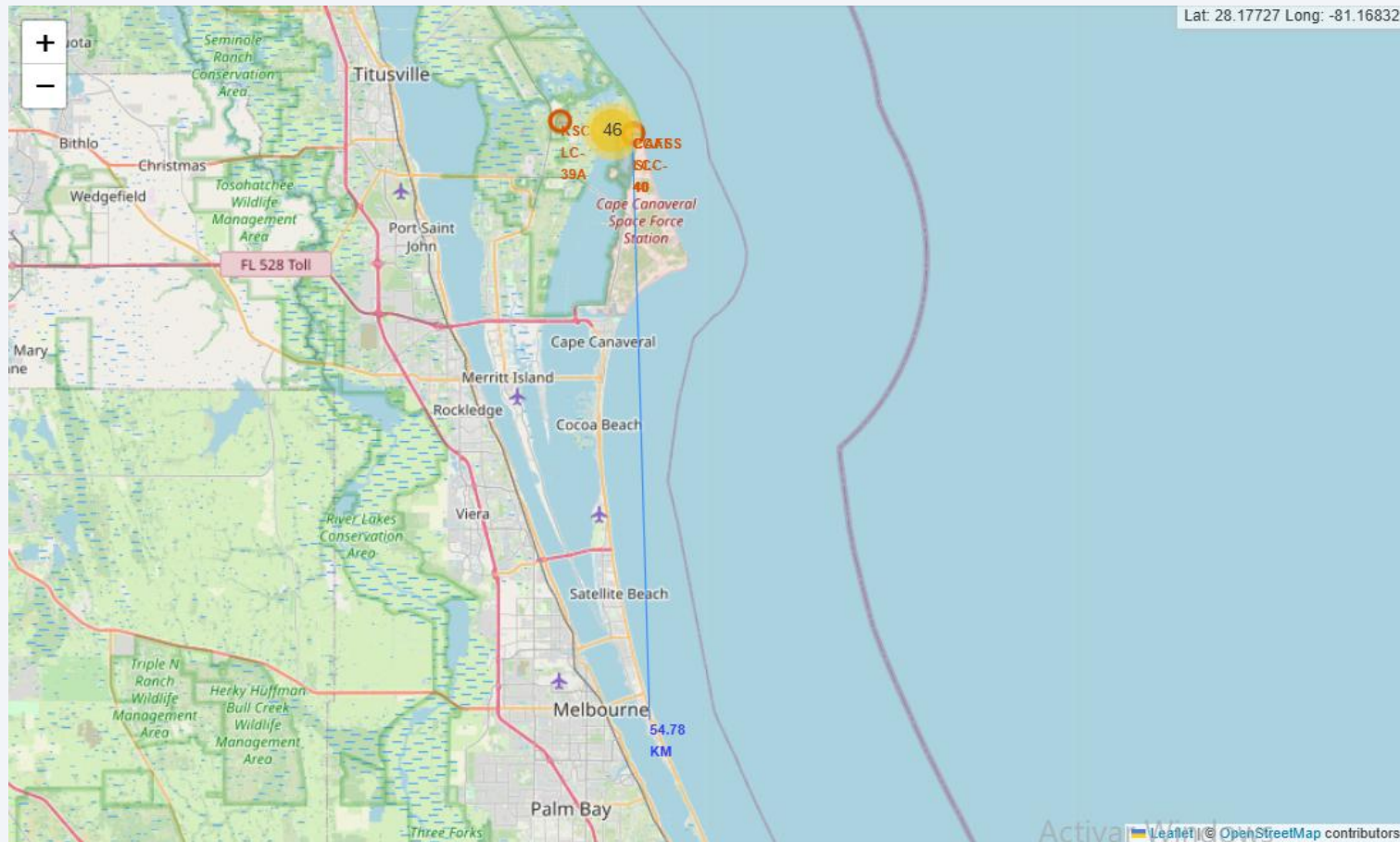
Folium Map – Success and Unsuccessful Launch Sites

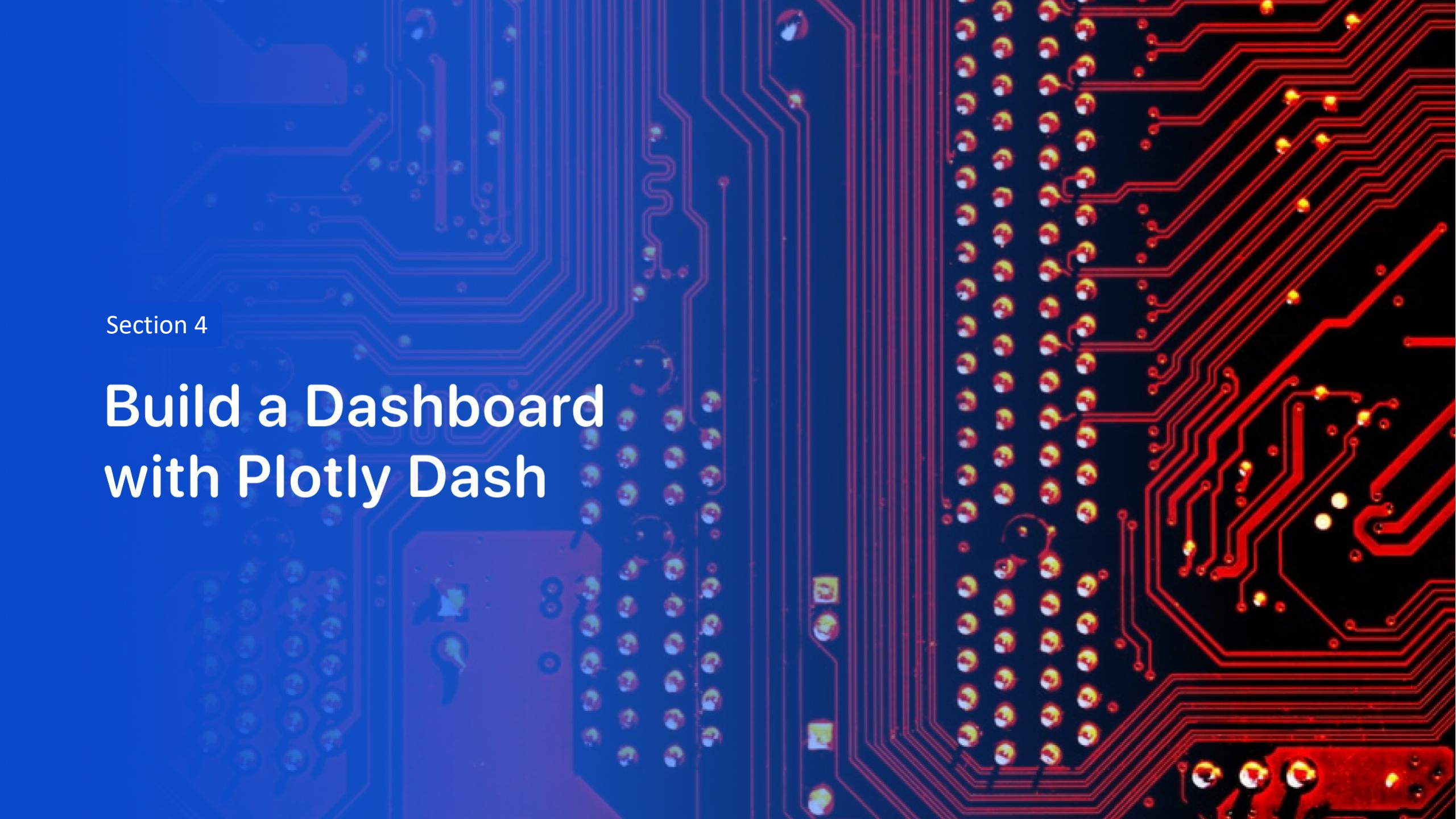
- The successful and unsuccessful Launch Sites. The green color shows successful Launch while red color shows unsuccessful Launch.



Folium Map – Distance from Launch Site to Melbourne

- Maps shows the distance between the Launch Site to Melbourne, Florida with a blue line.





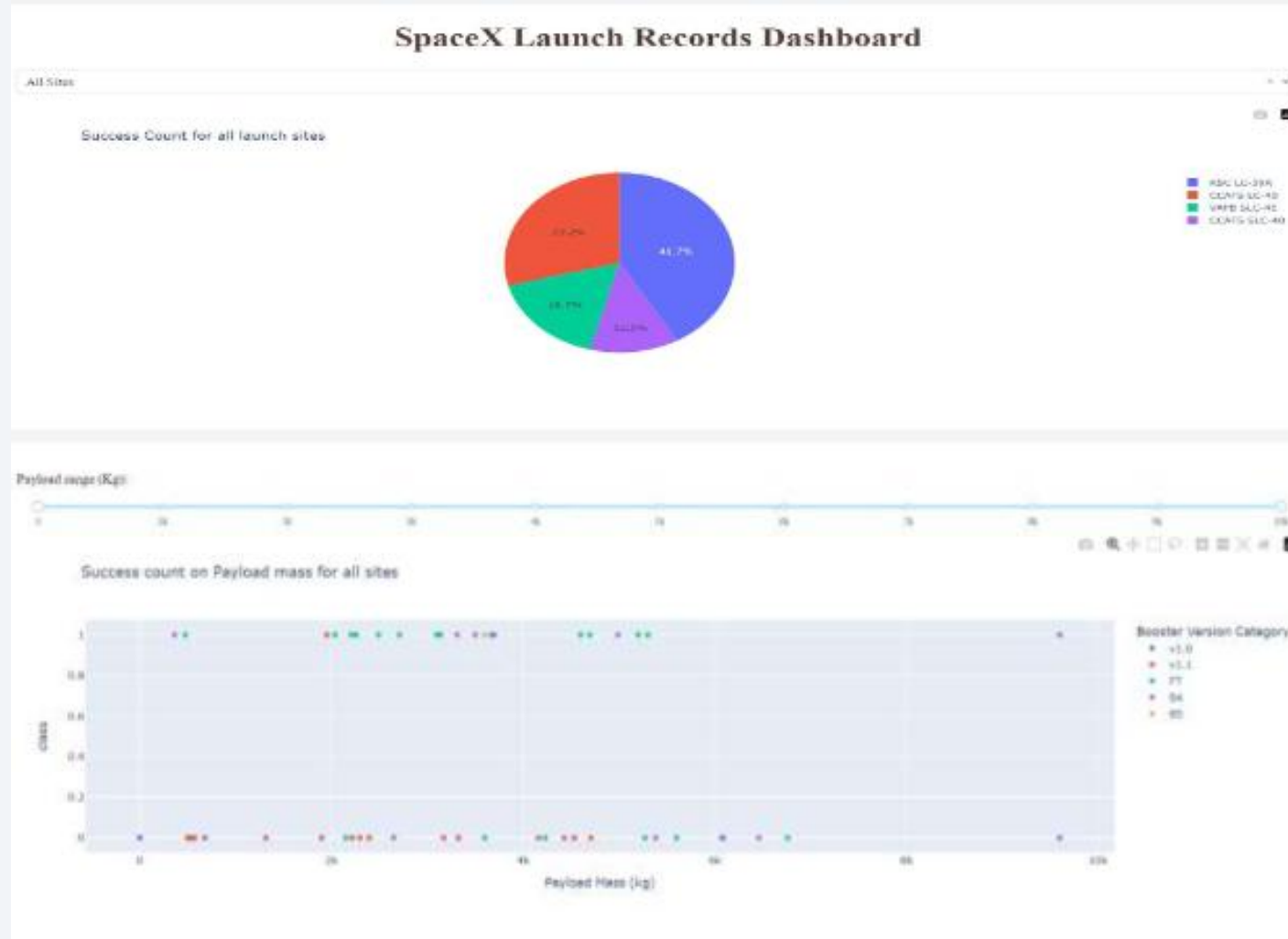
Section 4

Build a Dashboard with Plotly Dash

Plotly Dashboard

- Dashboard application built with Plotly to include:
 - Dropdown menu to select Launch Sites.
 - Pie chart displaying success rate for the selected Launch Site in the dropdown menu.
 - Scatter plot chart displaying the Payload Mass, Launch Site and Success/Failure.
 - Range slider for selecting range of Payload Mass in kg.

Plotly Dashboard



Section 5

Predictive Analysis (Classification)

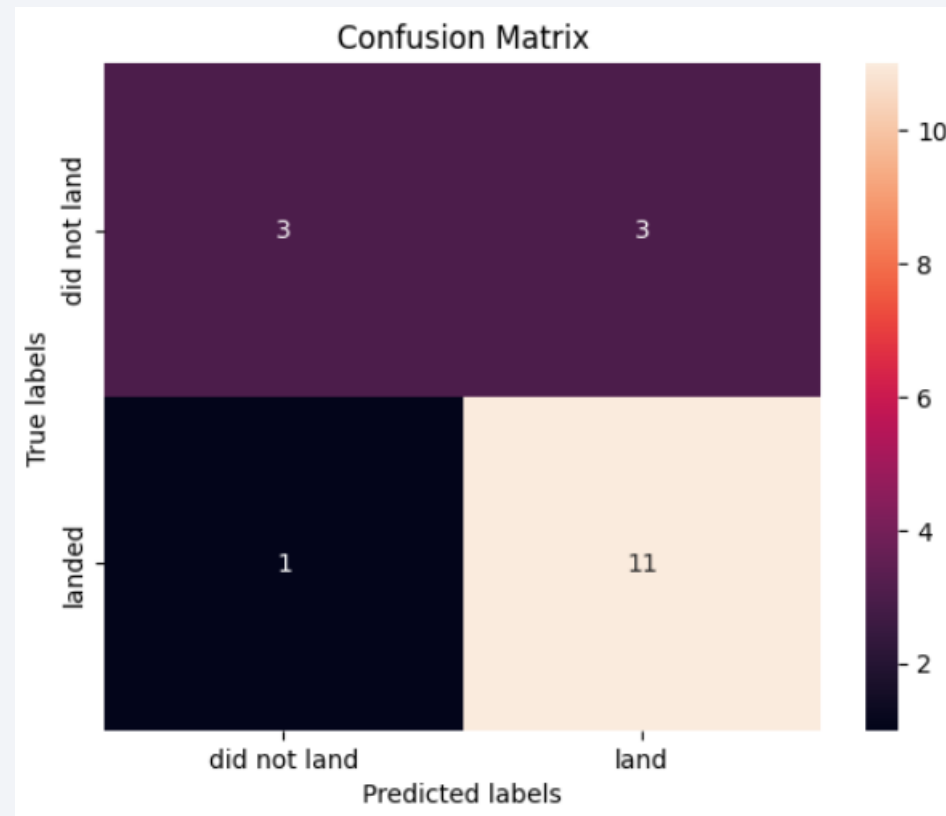
Classification Accuracy

- The final results for the Accuracy for each ML Model performed:
 - 4 predictive models (Logistic Regression, SVM, KNN & Decision Tree).
 - Decision tree model shows less accuracy of 0.89 against the rest that have 0.83 accuracy.

0	
Method	Test Data Accuracy
Logistic_Reg	0.833333
SVM	0.833333
Decision Tree	0.777778
KNN	0.833333

Confusion Matrix

- Decision Tree shows less accuracy from the rest of the predictive models.
- The Confusion Matrix shows 1 less correct prediction for the Landed-Did not Landed section than the rest of the models.



Conclusions

- The Launch Sites have different successful rate of landing; KSLC LC-39A and VAFB SLC-4E has success of 77%.
- When the Flight Number increases in each Launch Site, the success rate also increases.
- Orbits ES-L1, GEO, HEO and SSO have 100% of success rate compared to SO orbit that has 0% of success rate.
- In the Payload vs Launch Site Scatter plot, we can observe that there are no rockets launched for heavy payload mass in VAFB SLC launch site.
- For LEO orbit, the success is related to the number of flights and for GTO orbit, there seems to be no relationship between the flight number and the orbit.

Appendix

- https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/web_scraping.ipynb
- <https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>
- <https://github.com/betofrr/IBM-Data-Science-Certification/blob/main/edadataviz.ipynb>
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Thank you!

